

MORNING SESSION TASK

TASK-1 : DATABASE CREATION

```
mysql> create database college_db;
```

Query OK, 1 row affected (0.01 sec)

TASK-2 : TO DISPLAY DATABASES

```
mysql> show databases;
```

```
+-----+
| Database      |
+-----+
| college_db    |
| demo          |
| fruitmitra    |
| information_schema |
| mysql         |
| performance_schema |
| sample_test   |
| sweetydb      |
| sys           |
| trainingdb    |
| trainingdemo  |
+-----+
```

11 rows in set (0.00 sec)

SELECTING DB TO USE

```
mysql> use college_db;
```

Database changed

TASK-3 : TABLE CREATION

```
mysql> create table students(student_id INT, student_name VARCHAR(20), student_age INT,
gender VARCHAR(10), city VARCHAR(20));
```

Query OK, 0 rows affected (0.02 sec)

TASK-4: TO DISPLAY DATABASE SCHEMA

```
mysql> desc students;
```

Field	Type	Null	Key	Default	Extra
student_id	int	YES		NULL	
student_name	varchar(20)	YES		NULL	
student_age	int	YES		NULL	
gender	varchar(10)	YES		NULL	
city	varchar(20)	YES		NULL	

5 rows in set (0.00 sec)

TASK-5: ADDING EMAIL COLUMN TO STUDENT TABLE

```
mysql> alter table students add column email VARCHAR(20);
```

Query OK, 0 rows affected (0.05 sec)

Records: 0 Duplicates: 0 Warnings: 0

TO DISPLAY DATABASE SCHEMA

```
mysql> desc students;
```

Field	Type	Null	Key	Default	Extra
student_id	int	YES		NULL	
student_name	varchar(20)	YES		NULL	
student_age	int	YES		NULL	
gender	varchar(10)	YES		NULL	
city	varchar(20)	YES		NULL	
email	varchar(20)	YES		NULL	

```
+-----+-----+-----+-----+-----+
```

6 rows in set (0.00 sec)

TASK-6: ADDING PHONENUM COLUMN TO STUDENT TABLE

```
mysql> alter table students add column phoneNo INT(10);
```

Query OK, 0 rows affected, 1 warning (0.05 sec)

Records: 0 Duplicates: 0 Warnings: 1

TASK-7 : MODIFYING COLUMN

```
mysql> alter table students modify student_name VARCHAR(100);
```

Query OK, 0 rows affected (0.06 sec)

Records: 0 Duplicates: 0 Warnings: 0

TO DISPLAY DATABASE SCHEMA

```
mysql> desc students;
```

```
+-----+-----+-----+-----+-----+
```

```
| Field      | Type      | Null | Key | Default | Extra |
```

```
+-----+-----+-----+-----+-----+
```

```
| student_id | int       | YES  |     | NULL    |      |
```

```
| student_name | varchar(100) | YES  |     | NULL    |      |
```

```
| student_age | int       | YES  |     | NULL    |      |
```

```
| gender      | varchar(10) | YES  |     | NULL    |      |
```

```
| city        | varchar(20) | YES  |     | NULL    |      |
```

```
| email       | varchar(20) | YES  |     | NULL    |      |
```

```
| phoneNo     | int         | YES  |     | NULL    |      |
```

```
+-----+-----+-----+-----+-----+
```

7 rows in set (0.00 sec)

TASK-8 : RENAMING THE COLUMN

```
mysql> alter table students rename column city to student_city;
```

Query OK, 0 rows affected (0.02 sec)

Records: 0 Duplicates: 0 Warnings: 0

TO DISPLAY DATABASE SCHEMA

```
mysql> desc students;
```

Field	Type	Null	Key	Default	Extra
student_id	int	YES		NULL	
student_name	varchar(100)	YES		NULL	
student_age	int	YES		NULL	
gender	varchar(10)	YES		NULL	
student_city	varchar(20)	YES		NULL	
email	varchar(20)	YES		NULL	
phoneNo	int	YES		NULL	

```
7 rows in set (0.00 sec)
```

AFTERNOON SESSION

TASK

```
mysql> CREATE DATABASE practice;
```

```
Query OK, 1 row affected (0.01 sec)
```

```
mysql> use practice;
```

```
Database changed
```

```
mysql> CREATE TABLE students (
```

```
-> student_id INT,
```

```
-> student_name VARCHAR(50) NOT NULL,
```

```
-> email VARCHAR(100) NOT NULL
```

```
-> );
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> INSERT INTO students VALUES (1, 'Ramya', 'ramya@gmail.com');
```

Query OK, 1 row affected (0.01 sec)

```
mysql> insert into students values(2,'sweety',null);
```

ERROR 1048 (23000): Column 'email' cannot be null

```
mysql> alter table students add constraint unique_email UNIQUE (email);
```

Query OK, 0 rows affected (0.07 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
mysql> insert into students values(3,'priya','priya@gmail.com');
```

Query OK, 1 row affected (0.01 sec)

```
mysql> insert into students values(4,'siri','siri@gmail.com');
```

Query OK, 1 row affected (0.01 sec)

```
mysql> insert into students values(4,'siri','priya@gmail.com');
```

ERROR 1062 (23000): Duplicate entry 'priya@gmail.com' for key 'students.unique_email'

```
mysql> desc students;
```

Field	Type	Null	Key	Default	Extra
student_id	int	YES		NULL	
student_name	varchar(50)	NO		NULL	
email	varchar(100)	NO	PRI	NULL	

3 rows in set (0.02 sec)

```
mysql> alter table students modify student_id int PRIMARY KEY;
```

Query OK, 0 rows affected (0.11 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
mysql> desc students;
```

Field	Type	Null	Key	Default	Extra
student_id	int	NO	PRI	NULL	
student_name	varchar(50)	NO		NULL	
email	varchar(100)	NO	UNI	NULL	

3 rows in set (0.00 sec)

```
mysql> insert into students values(5,'a','a@gmail.com');
```

Query OK, 1 row affected (0.01 sec)

```
mysql> insert into students values(6,'b','b@gmail.com');
```

Query OK, 1 row affected (0.01 sec)

```
mysql> insert into students values(5,'c','c@gmail.com');
```

ERROR 1062 (23000): Duplicate entry '5' for key 'students.PRIMARY'

```
mysql> alter table students modify student_id INT AUTO_INCREMENT;
```

Query OK, 5 rows affected (0.06 sec)

Records: 5 Duplicates: 0 Warnings: 0

```
mysql> insert into students values('m','m@gmail.com');
```

ERROR 1136 (21S01): Column count doesn't match value count at row 1

```
mysql> insert into students values(null,'m@gmail.com');
```

ERROR 1136 (21S01): Column count doesn't match value count at row 1

```
mysql> insert into students (student_name,email) values('l',l
```

```
-> ;
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near " at line 1

```
mysql> insert into students (student_name,email) values('l','l@gmail.com');
```

Query OK, 1 row affected (0.01 sec)

```
mysql> insert into students (student_name,email) values('v','v@gmail.com');
```

Query OK, 1 row affected (0.01 sec)

```
mysql> select * from students;
```

```
+-----+-----+-----+
| student_id | student_name | email      |
+-----+-----+-----+
| 1 | Ramya    | ramya@gmail.com |
| 3 | priya    | priya@gmail.com |
| 4 | siri     | siri@gmail.com  |
| 5 | a        | a@gmail.com     |
| 6 | b        | b@gmail.com     |
| 7 | l        | l@gmail.com     |
| 8 | v        | v@gmail.com     |
+-----+-----+-----+
```

7 rows in set (0.00 sec)

```
mysql> alter table students add city varchar(50) default 'mpl';
```

Query OK, 0 rows affected (0.01 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
mysql> insert into students (student_name,email) values('k','k@gmail.com');
```

Query OK, 1 row affected (0.01 sec)

```
mysql> select * from students;
```

student_id	student_name	email	city
1	Ramya	ramya@gmail.com	mpl
3	priya	priya@gmail.com	mpl
4	siri	siri@gmail.com	mpl
5	a	a@gmail.com	mpl
6	b	b@gmail.com	mpl
7	l	l@gmail.com	mpl
8	v	v@gmail.com	mpl
9	k	k@gmail.com	mpl

8 rows in set (0.00 sec)

```
mysql> alter table students add column age int;
```

Query OK, 0 rows affected (0.01 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
mysql> desc students;
```

Field	Type	Null	Key	Default	Extra
-------	------	------	-----	---------	-------

```

+-----+-----+-----+-----+
| student_id | int      | NO  | PRI | NULL | auto_increment |
| student_name | varchar(50) | NO  |     | NULL |                |
| email       | varchar(100) | NO  | UNI | NULL |                |
| city       | varchar(50) | YES |     | mpl  |                |
| age        | int        | YES |     | NULL |                |
+-----+-----+-----+-----+

```

5 rows in set (0.00 sec)

```
mysql> alter table students constraint check_age(age>=18);
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'constraint check_age(age>=18)' at line 1

```
mysql> alter table students add constraint check_age(age>=18);
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '(age>=18)' at line 1

```
mysql> alter table students add constraint check_age check (age>=18);
```

Query OK, 8 rows affected (0.08 sec)

Records: 8 Duplicates: 0 Warnings: 0

```
mysql> desc students;
```

```

+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra          |
+-----+-----+-----+-----+
| student_id | int       | NO   | PRI | NULL    | auto_increment |
| student_name | varchar(50) | NO   |     | NULL    |                |
| email      | varchar(100) | NO   | UNI | NULL    |                |
| city      | varchar(50) | YES  |     | mpl     |                |

```

age	int	YES		NULL	
-----	-----	-----	-----	-----	-----

5 rows in set (0.00 sec)

```
mysql> insert into students (student_name,email,age) values('d','d@gmail.com',20);
```

Query OK, 1 row affected (0.01 sec)

```
mysql> insert into students (student_name,email,age) values('d','d@gmail.com',13);
```

ERROR 3819 (HY000): Check constraint 'check_age' is violated.

```
mysql> create table departments ( dept_id int primary key, dept_name varchar(50) );
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> insert into departments values(101,'CAI');
```

Query OK, 1 row affected (0.01 sec)

```
mysql> insert into departments values(102,'CSE');
```

Query OK, 1 row affected (0.01 sec)

```
mysql> insert into students (student_name,email,age,dept_id) values
('t','t@gmail.com',21,101);
```

ERROR 1054 (42S22): Unknown column 'dept_id' in 'field list'

```
mysql> alter table students add dept
```

```
-> _id int;
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '_id int' at line 2

```
mysql> alter table students add dept_id int;
```

Query OK, 0 rows affected (0.02 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
mysql> alter table students add constarint fk_department foreign key (dept_id) references departments(dept_id);
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'fk_department foreign key (dept_id) references departments(dept_id)' at line 1

```
mysql> ALTER TABLE students
```

```
-> ADD CONSTRAINT fk_department
```

```
-> FOREIGN KEY (dept_id)
```

```
-> REFERENCES departments(dept_id);
```

Query OK, 9 rows affected (0.07 sec)

Records: 9 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO departments VALUES (101, 'CSE');
```

ERROR 1062 (23000): Duplicate entry '101' for key 'departments.PRIMARY'

```
mysql> truncate table departments;
```

ERROR 1701 (42000): Cannot truncate a table referenced in a foreign key constraint (`practice`.`students`, CONSTRAINT `fk\_department`)

```
mysql> INSERT INTO departments VALUES (1, 'CSC');
```

Query OK, 1 row affected (0.01 sec)

```
mysql> INSERT INTO departments VALUES (2, 'ECE');
```

Query OK, 1 row affected (0.01 sec)

```
mysql> insert into students (student_name,email,age,dept_id) values ('t','t@gmail.com',21,1);
```

Query OK, 1 row affected (0.01 sec)

```
mysql> insert into students (student_name,email,age,dept_id) values ('n','n@gmail.com',22,88);
```

ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails
(`practice`.`students`, CONSTRAINT `fk\_department` FOREIGN KEY (`dept\_id`) REFERENCES
`departments` (`dept\_id`))

mysql>